

TROUBLESHOOTING TOOLS AND MASTERY

1. Use the ping command to check the connectivity between your computer and www.spotify.com. Note the response times and explain what they indicate about your network connection.

-> **Run the ping command**

On **Windows**, open **Command Prompt** and enter:

```
ping www.spotify.com
```

You may get output similar to:

Pinging www.spotify.com [xxx.xxx.xxx.xxx] with 32 bytes of data:

Reply from xxx.xxx.xxx.xxx: bytes=32 time=25ms TTL=...

Reply from xxx.xxx.xxx.xxx: bytes=32 time=27ms TTL=...

Reply from xxx.xxx.xxx.xxx: bytes=32 time=24ms TTL=...

Reply from xxx.xxx.xxx.xxx: bytes=32 time=26ms TTL=...

Ping statistics for xxx.xxx.xxx.xxx:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 24ms, Maximum = 27ms, Average = 25ms

Use the actual values displayed on your computer in your assignment. The numbers above are only an example.

2. Record the results

Test	Result
Website	www.spotify.com
Packets sent	4

Test	Result
Packets received	Record actual result
Packet loss	Record actual %
Minimum time	Record actual ms
Maximum time	Record actual ms
Average time	Record actual ms

3. What does response time mean?

The time value is the approximate **round-trip time (RTT)** for the ICMP packet to reach the destination and for the response to return.

- **Low and consistent times:** Generally indicate a responsive connection with relatively little delay.
- **Higher times:** Indicate greater network latency, which can result from distance, routing, congestion, or other network conditions.
- **Large variations (jitter):** May indicate changing network conditions or congestion.
- **Request timed out:** Means a response was not received; this does **not necessarily mean your Internet is down**, because some servers or networks may not respond to ICMP ping requests.

Assignment conclusion

I used the ping `www.spotify.com` command to test connectivity from my computer to Spotify. The test returned an average response time of ___ **ms**, with ___% **packet loss**. The response time represents the round-trip delay between my computer and the destination. A low, consistent response time indicates relatively low network latency, while higher or fluctuating times indicate greater network delay. Spotify's web service is available through its official website.

2. Your friend is unable to access `www.zomato.com`, but other websites work fine. Use `nslookup` to check the DNS resolution for `www.zomato.com` and describe what the output tells you.

Hint: Look for the IP address and any error messages in the `nslookup` result.

->Objective

To use the `nslookup` command to check whether `www.zomato.com` can be resolved to an IP address and understand what the result indicates about the DNS configuration.

Zomato's official website is available at www.zomato.com.

Step 1: Open Command Prompt

On Windows:

1. Press **Windows + R**.
2. Type cmd.
3. Press **Enter**.

Step 2: Run nslookup

Type:

```
nslookup www.zomato.com
```

Press **Enter**.

Example of successful output

The exact IP address may vary because websites can use multiple servers or DNS/CDN infrastructure.

```
C:\>nslookup www.zomato.com
```

```
Server: dns.google
```

```
Address: 8.8.8.8
```

```
Non-authoritative answer:
```

```
Name: www.zomato.com
```

```
Addresses: xxx.xxx.xxx.xxx
```

Step 3: Interpret the result

Output	Meaning
Server	Shows which DNS server answered the query.
Address under Server	IP address of the DNS server being used.
Name: www.zomato.com	Confirms that the domain name was found.

Output	Meaning
Addresses:	Shows the IP address(es) associated with the domain.
Non-authoritative answer	The response came from a DNS resolver/cache rather than directly from the domain's authoritative DNS server.

If an IP address is returned

If nslookup returns an IP address, **DNS resolution is working**. Therefore, if your friend still cannot open Zomato while other websites work, the problem may be elsewhere, such as browser settings, firewall/security software, routing, or the website connection.

If an error appears

For example:

DNS request timed out.

*** Request to dns.google timed-out

or:

*** Can't find www.zomato.com: Non-existent domain

this indicates a **DNS resolution problem**. The computer is unable to obtain a valid IP address for the domain.

Conclusion

nslookup helps determine whether the problem is related to DNS. If www.zomato.com resolves to an IP address, DNS is successfully translating the domain name into an IP address, so troubleshooting should continue beyond DNS. If nslookup returns an error or cannot find the domain, the DNS configuration or DNS server should be investigated.

Screenshot to submit: Capture the Command Prompt showing the complete nslookup www.zomato.com command and its output.

3. Open Wireshark and capture network traffic while you refresh your Instagram feed in a browser. Identify and list the different protocols you see in the capture.

***Hint:** Filter by protocol type (like HTTP, TCP, DNS) in Wireshark to make it easier.*

-> Objective

To capture network traffic in Wireshark while refreshing an Instagram feed and identify the protocols involved in loading the webpage.

Step 1: Start Wireshark

1. Open **Wireshark**.
2. Select the active network interface:
 - **Wi-Fi** if connected wirelessly.
 - **Ethernet** if using a LAN cable.
3. Click **Start Capturing Packets**.
4. Open your browser and visit Instagram.
5. Refresh the Instagram feed using **Ctrl + R**.
6. Wait for the feed and images/videos to load.
7. Return to Wireshark and click the **Stop** button.

Wireshark's display filters allow packets to be filtered by protocol without deleting other packets from the capture.

Step 2: Apply Protocol Filters

Use the Wireshark display-filter box and try these filters one at a time:

Filter	Protocol	What to observe
dns	DNS	Domain-name lookups
tcp	TCP	Reliable transport connections
udp	UDP	UDP-based traffic
tls	TLS	Encrypted HTTPS communication
http	HTTP	Unencrypted HTTP traffic, if any
quic	QUIC	QUIC traffic, if used by the browser/service
ip	IPv4	IP packets
ipv6	IPv6	IPv6 packets

For example, entering tcp in the display-filter bar shows packets containing TCP.

Step 3: Record the Protocols

A typical Instagram browser capture may contain protocols such as:

Protocol	Purpose
DNS	Resolves Instagram-related domain names into IP addresses.
TCP	Provides reliable transport for connections that use TCP.
TLS	Encrypts HTTPS application traffic.
UDP	Carries UDP-based network traffic.
QUIC	A modern transport protocol that runs over UDP and may be used by browsers/services.
IPv4/IPv6	Provides addressing and routing between your computer and Internet servers.
Ethernet/Wi-Fi	Handles local network communication between your device and network equipment.

You should **record only the protocols that actually appear in your own capture**. The exact list can vary depending on the browser, operating system, network, DNS configuration, and Instagram's current infrastructure. Wireshark's documentation specifically supports filtering and inspecting protocols such as TCP, TLS, HTTP, HTTP/2 and QUIC.

Important observation

You may **not see ordinary HTTP packets** when loading Instagram. Modern websites normally use encrypted HTTPS connections, so you are more likely to see **TLS** or possibly **QUIC** rather than readable HTTP requests.

Screenshot to Submit

Take a screenshot showing:

1. Wireshark packet list after refreshing Instagram.
2. The display-filter bar.
3. Several visible protocol names in the **Protocol** column.
4. Optionally, another screenshot with dns or tls entered in the filter bar.

Result

The Wireshark capture demonstrates that loading an Instagram feed involves multiple networking protocols. **DNS** can be used for domain resolution, while **TCP/TLS or QUIC** can carry the encrypted communication between the browser and Internet services.

The exact protocols observed should be recorded from the student's actual packet capture.

4. A user reports that their Flipkart app is slow to load product images. Use netstat to list all current network connections on your system and identify if there are any suspicious or high-usage connections that could be affecting bandwidth.

-> **Objective**

To use the netstat command to list current network connections and identify connections that may be using network resources and affecting the loading speed of Flipkart product images.

Step 1: Open Command Prompt

1. Press **Windows + R**.
2. Type cmd.
3. Press **Enter**.
4. For more detailed information, open **Command Prompt as Administrator**.

Step 2: List Current Connections

Run:

```
netstat -ano
```

The -a option displays active connections and listening ports, -n shows numerical IP addresses and ports, and -o adds the process ID (PID) associated with each connection.

Example:

Active Connections

Proto	Local Address	Foreign Address	State	PID
TCP	192.168.1.5:52341	142.250.183.14:443	ESTABLISHED	4520
TCP	192.168.1.5:52345	104.18.25.10:443	ESTABLISHED	6284
TCP	192.168.1.5:52350	13.107.42.12:443	ESTABLISHED	3176

Do not use these IP addresses as your actual results. Your output will be different.

Step 3: Identify the Applications

Note the **PID** of any connection that looks unusual. Then open **Task Manager → Details** and match the PID with the running process.

You can also use:

```
netstat -abno
```

This can show the executable associated with connections, although it may require administrator privileges and can take longer to run.

Step 4: Look for Possible Bandwidth-Using Connections

Look for:

- Many ESTABLISHED connections from applications you are not currently using.
- Large numbers of connections belonging to cloud-storage or file-sync applications.
- Game launchers downloading updates.
- Video/music streaming applications.
- Unknown applications making repeated external connections.
- Unexpected listening ports or processes.

However, **netstat does not directly show how many MB/s each connection is using**.

It shows connections, addresses, ports, states and PIDs. Use **Task Manager → Processes/Performance** or Resource Monitor to determine actual bandwidth consumption.

Step 5: Check Network Statistics

You can also run:

```
netstat -e
```

This displays Ethernet statistics, including packets and bytes sent and received.

For continuously changing connections:

```
netstat -ano 5
```

This refreshes the connection list every 5 seconds. Press **Ctrl + C** to stop it.

Analysis

Observation	Possible meaning
Few normal ESTABLISHED connections	Usually normal
Many connections from a browser	Could be normal when multiple websites/services are open
Cloud-sync application transferring data	Could consume significant bandwidth
Game/software update running	Could slow other Internet traffic
Unknown process with many connections	Investigate the process and PID
High network usage in Task Manager	Another application may be competing for bandwidth
TIME_WAIT connections	Often normal after connections have recently closed

Important point

An unfamiliar IP address **does not automatically mean it is suspicious**. Modern applications frequently connect to content-delivery networks, cloud providers, advertising services and other third-party infrastructure.

Also, netstat alone cannot prove that a connection is consuming high bandwidth. The actual traffic rate should be checked with Task Manager or Resource Monitor.

Conclusion

The netstat -ano command provides a list of current network connections and their associated PIDs. By matching unusual connections to applications and then checking network usage in Task Manager/Resource Monitor, we can determine whether another application is consuming bandwidth that could contribute to slow loading of Flipkart product images.

Screenshots to Submit

1. Command Prompt showing:
2. netstat -ano
3. The PID of any connection you investigated.
4. **Task Manager → Details** showing the matching application.

5. **Task Manager → Performance → Ethernet/Wi-Fi** showing network activity.

Expected result: Record your actual connections and identify whether any background application is generating significant network traffic. Do not invent IP addresses, PIDs, or bandwidth values in the lab report.

5. Your college WiFi keeps disconnecting during IPL match streaming. Create a troubleshooting checklist using ping, tracert, and netstat to systematically diagnose the issue step by step. For each tool, write what command you would use and what result would indicate a problem.

-> **Wi-Fi Troubleshooting Checklist**

Problem

The college Wi-Fi frequently disconnects while streaming an IPL match. The following three Windows commands can be used systematically to determine whether the problem is with the **local Wi-Fi connection, routing/Internet path, or network activity on the computer**.

Step 1: Test the Local Network with ping

First, find the default gateway by running:

ipconfig

Look for **Default Gateway**, for example:

Default Gateway : 192.168.1.1

Then ping the gateway:

ping 192.168.1.1

You can also test Internet connectivity:

ping 8.8.8.8

The ping command sends ICMP Echo Requests and reports whether replies are received and their round-trip time.

What indicates a problem?

Result	Meaning
Consistent replies with low latency	Local connection is probably working normally

Result	Meaning
Request timed out	Possible Wi-Fi connectivity or packet-loss problem
Very high/unstable response times	Possible congestion, interference, or weak Wi-Fi
Gateway ping fails	Problem is likely between the computer and college Wi-Fi/router
Gateway works but 8.8.8.8 fails	Possible Internet/WAN or routing problem

Important: A failed ping does not always prove that the network is down because some devices or networks may block ICMP traffic.

Step 2: Find Where the Problem Occurs with tracert

Run:

tracert 8.8.8.8

Or trace a website:

tracert www.google.com

tracert shows the sequence of routers/hops between your computer and the destination and reports round-trip times for responding hops.

Example:

Tracing route to 8.8.8.8

```

1  2 ms  3 ms  2 ms  192.168.1.1
2  10 ms 12 ms 11 ms ...
3  25 ms 27 ms 26 ms ...
4  30 ms 31 ms 29 ms 8.8.8.8

```

What indicates a problem?

Observation	Possible problem
First hop has high/variable latency	Local Wi-Fi/router congestion
A later hop suddenly becomes very slow	Possible congestion or routing issue beyond the local network
Several consecutive * * * hops	Router may be filtering/non-responding to traceroute; investigate further rather than assuming failure
Destination net unreachable	Routing problem
Destination cannot be reached	Possible Internet path or destination issue

Asterisks alone do **not** necessarily indicate a failure because some routers do not return the ICMP messages used by tracert.

Step 3: Check Active Connections with netstat

Run:

```
netstat -ano
```

This displays active connections, remote addresses, connection states and the associated process IDs (PIDs).

For continuously changing connections:

```
netstat -ano 5
```

This refreshes the results every 5 seconds.

You can also check Ethernet statistics:

```
netstat -e
```

What indicates a problem?

Look for:

- A large number of connections from applications you are not using.
- Cloud-storage synchronization running during the match.
- Game/software downloads.

- Video or other streaming applications consuming network resources.
- Unknown applications with many active connections.

Match a suspicious PID with **Task Manager** → **Details** to identify the application.

Note: netstat shows connections and statistics, but it does not directly tell you which connection is consuming the most bandwidth. Use Task Manager or Resource Monitor for actual bandwidth usage.

Step 4: Troubleshooting Decision Table

Test	Normal result	Problem result	Next action
ping <gateway>	Stable replies	Timeout/high latency	Check Wi-Fi signal, access point and local congestion
ping 8.8.8.8	Stable replies	Timeout/loss	Investigate Internet connection
tracert 8.8.8.8	Normal progression	High delay/unreachable hop	Identify where the path becomes problematic
netstat -ano	Normal application connections	Many unexpected connections	Identify processes using the network
netstat -e	Normal traffic statistics	Rapidly increasing traffic	Check for downloads/background activity

Step 5: Record the Results

Use this table in the lab report:

Command	Observation	Problem?
ping <gateway>	_____	Yes / No
ping 8.8.8.8	_____	Yes / No
tracert 8.8.8.8	_____	Yes / No
netstat -ano	_____	Yes / No

Command	Observation	Problem?
netstat -e	_____	Yes / No

Conclusion

The troubleshooting should start with **ping** to determine whether the computer can reliably communicate with the local gateway and Internet. Next, **tracert** helps identify where delays or routing problems occur. Finally, **netstat** helps identify active connections and processes that may be contributing to network activity.

If the **gateway ping itself is unstable**, the college Wi-Fi/local network should be investigated first. If the gateway is stable but later hops show problems, the issue may be farther along the Internet path. If connectivity is normal but streaming still buffers, checking bandwidth usage and application connections with **netstat plus Task Manager/Resource Monitor** is appropriate.